

When $A = 36 \text{ km/L}$

$$v^2 = \frac{22500}{36} \Rightarrow v = \sqrt{\frac{22500}{36}} = \frac{150}{6}$$

Hence, the speed of a car is 25 km/h .

17. (c) Let the length of each train be $I \text{ m}$.

$$\Rightarrow \text{Speed of first train} = \left(\frac{I}{4}\right) \text{ m/s}$$

$$\text{and speed of second train} = \left(\frac{I}{5}\right) \text{ m/s}$$

As, both trains are moving in opposite direction.

Time taken to cross each other

$$= \frac{I + I}{\frac{I}{4} + \frac{I}{5}} \quad [\text{Rule 7}]$$

$$= \left(\frac{2I}{\frac{9I}{20}}\right) \text{ s} = \left(\frac{20 \times 2}{9}\right) = \frac{40}{9} \text{ s}$$

18. (b) Given, total distance to cover in 10 h
 $= 80 \text{ km}$

If it covers 40 km in $\frac{3}{5}$ th of time i.e.
 40 km in 6 h .

$$\therefore \text{Remaining time} = 10 - 6 = 4 \text{ h}$$

$$\text{Remaining distance} = 40 \text{ km}$$

$$\therefore \text{Required speed} = (40 \div 4) \text{ km/h}$$

$$= \frac{40}{4} = 10 \text{ km/h}$$

19. (c) Here, $x = 3 \text{ m/s}$, $y = 4 \text{ m/s}$
 and $t = 25 \text{ min}$

$$\therefore \text{Required time} = \frac{x \times t}{y - x} \quad [\text{Rule 3}]$$

$$= \frac{3 \times 25}{4 - 3} = 75 \text{ min}$$

20. (d) Distance travelled in $1 \text{ h} = 48 \text{ km}$

$$\therefore \text{Distance travelled in } 50 \text{ min}$$

$$= \frac{48}{60} \times 50 = 40 \text{ km}$$

$$\text{Time to be reduced} = \frac{40}{60} \text{ h}$$

$$\therefore \text{Required speed}$$

$$= \frac{40}{40/60} = \frac{40 \times 60}{40} = 60 \text{ km/h}$$

21. (a) Let the length of faster train be $x \text{ km}$.

$\therefore$ Trains travel in the same direction.

$$\therefore \text{Relative speed} = (50 - 32) = 18 \text{ km/h}$$

$$\text{Elapsed time} = 15 \text{ s} = \frac{15}{3600} \text{ h}$$

$$\text{Now, time} = \frac{\text{distance}}{\text{speed}}$$

$$\Rightarrow \frac{15}{3600} = \frac{x}{18} \Rightarrow x \times 3600 = 18 \times 15$$

$$\therefore x = \frac{18 \times 15}{3600} = 0.075 \text{ km} = 75 \text{ m}$$

22. (b) Let distance between A and B be $x \text{ km}$.

By given condition,

$$\frac{x}{12} - \frac{x}{18} = 2 \Rightarrow 6x = 2 \times 18 \times 12$$

$$\therefore x = \frac{2 \times 18 \times 12}{6} = 72 \text{ km}$$

Hence, the required distance is 72 km .

23. (b) Relative speed when trains are in opposite direction

$$= V_1 + V_2$$

$$= 75 + 50 = 125 \text{ km/h}$$

$$= \frac{125 \times 5}{18} \text{ m/s}$$

and total distance covered

$$= (100 + 150) = 250 \text{ m}$$

$\therefore$ Time taken to cross each other

$$= \frac{\text{Total covered distance}}{\text{Relative speed}}$$

$$= \frac{250 \times 18}{125 \times 5} = 7.2 \text{ s}$$

24. (d) Let he walked up the hill at a distance of $x \text{ km}$.

By given condition,

$$\frac{x}{2.5} + \frac{x}{3.25} = 4 \frac{36}{60}$$

$$\Rightarrow x \left(\frac{1}{2.5} + \frac{1}{3.25} \right) = \frac{276}{60}$$

$$\Rightarrow x \left(\frac{2}{5} + \frac{4}{13} \right) = \frac{276}{60}$$

$$\therefore x = \frac{276}{60} \times \frac{65}{46} = 6.5 \text{ km}$$

25. (b) Let speed of boat in still water

$$= x \text{ km/h}$$

and speed of boat in current water

$$= y \text{ km/h}$$

$$\therefore \text{Downstream speed} = (x + y) \text{ km/h}$$

$$\text{Upstream speed} = (x - y) \text{ km/h} \quad [\text{Rule 9}]$$

According to the question,

$$\frac{30}{x - y} + \frac{44}{x + y} = 10 \quad \dots(i)$$

$$\text{and } \frac{40}{x - y} + \frac{55}{x + y} = 13 \quad \dots(ii)$$

$$\text{Let } \frac{1}{x - y} = u \text{ and } \frac{1}{x + y} = v \quad \dots(iii)$$

$$\text{Then, } 30u + 44v = 10$$

$$40u + 55v = 13$$

On solving the above equations, we get

$$u = \frac{1}{5}, v = \frac{1}{11}$$

From Eq. (iii), we get

$$x - y = 5 \text{ and } x + y = 11$$

On solving the above equations, we get

$$x = 8, y = 3$$

Thus, the speed of boat in still water

$$= 8 \text{ km/h}$$

26. (a) Let the speed of motorboat be $x \text{ km/h}$ and the speed of water be $y \text{ km/h}$.

Now, speed of boat downstream

$$= (x + y) \text{ km/h} \quad [\text{Rule 9}]$$

and speed of boat upstream

$$= (x - y) \text{ km/h}$$

By given condition,

$$\frac{9}{x + y} = 2 \Rightarrow 2x + 2y = 9 \dots(i)$$

$$\text{and } \frac{9}{x - y} = 6 \Rightarrow 6x - 6y = 9$$

$$\Rightarrow 2x - 2y = 3 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$x = 3 \text{ km/h and } y = 1.5 \text{ km/h}$$

$$\therefore \text{Speed of boat} = 3 \text{ km/h}$$

27. (d) Let the distance between P and $Q = d \text{ km}$. Total time taken by Ram

$$= \frac{d}{10} + \frac{d}{15} = \frac{25d}{150}$$

$$\text{Total time taken by Shyam} = \frac{2d}{12.5} = \frac{4d}{25}$$

According to question,

$$\frac{25d}{150} - \frac{4d}{25} = \frac{12}{60}$$

$$\Rightarrow d \left[\frac{25 - 24}{150} \right] = \frac{1}{5}$$

$$\Rightarrow \frac{d}{150} = \frac{1}{5} \Rightarrow d = 30 \text{ km}$$

28. (a) Distance travelled by x in 15 min

$$= 60 \times \frac{15}{60} = 15 \text{ km}$$

Distance travelled by y in 10 min

$$= 48 \times \frac{10}{60} = 8 \text{ km}$$

$$\text{Difference} = (15 - 8) \text{ km} = 7 \text{ km}$$

Hence, at 3:15 pm they are 7 km apart.

So, statement I is true. As speed of x is greater than y . So, y will never overtake x .

Thus, statement II is false.

29. (c) I. Here, $x = 20 \text{ km/h}$, $y = 4 \text{ km/h}$,

$$t_1 = 30 \text{ min}, t_2 = 10 \text{ min}$$

According to formula,

$$\therefore \text{Required distance}$$

$$= (t_1 - t_2)(x + y) \frac{x}{y} \quad [\text{Rule 5}]$$

$$= \frac{(30 - 10)}{60} (20 + 4) \left(\frac{20}{4} \right)$$

$$= \frac{20}{60} \times 24 \times \frac{20}{4} = 5 \times 8 = 40 \text{ km}$$

So, I is incorrect.

II. Here, $x = 20$ km/h, $y = 10$ km/h
 $t_1 = 30$ min, $t_2 = 10$ min

According to formula,

∴ Required distance

$$= \left(\frac{30 - 10}{60} \right) (20 + 10) \left(\frac{20}{10} \right)$$

$$= \frac{20}{60} \times 30 \times \frac{20}{10} = 20 \text{ km}$$

So, II is correct.

30. (a) $x = 16$, $y = 25$

According to formula,

$$\therefore \frac{\text{Ramesh's Speed}}{\text{Prateek's Speed}} = \sqrt{\frac{y}{x}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

or 5:4

31. (a) Here, we know the relation

$$\frac{\text{Speed of Ramesh}}{\text{Speed of Prateek}} = \sqrt{\frac{y}{x}} \Rightarrow \frac{7}{9} = \sqrt{\frac{y}{x}}$$

On squaring both sides, $\frac{49}{81} = \frac{y}{x} = 81:49$

32. (d) To cover a distance by train

$$= 280 + 220 = 500 \text{ m}$$

Speed of train = 60 km/h

$$= 60 \times \frac{5}{18} \text{ m/s}$$

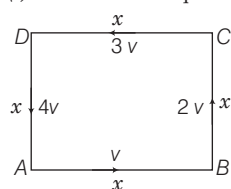
$$= \frac{50}{3} \text{ m/s}$$

∴ Time taken by train

$$= \frac{\text{total distance}}{\text{speed}}$$

$$= \frac{500}{50/3} = 30 \text{ s}$$

33. (c) Let side of a square be x .



∴ Average speed (u) = $\frac{\text{total distance}}{\text{total time}}$

$$= \frac{(x + x + x + x)}{\frac{x}{v} + \frac{x}{2v} + \frac{x}{3v} + \frac{x}{4v}}$$

$$= \frac{4x}{x \left(\frac{1}{v} + \frac{1}{2v} + \frac{1}{3v} + \frac{1}{4v} \right)}$$

$$= \frac{4 \times v}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{48v}{25} = 1.92 v$$

which lies in the interval $v < u < 2v$.

34. (d) Given, speed of a car = 45 km/h
 Time taken by the car

$$= 140 \text{ min} = \frac{140}{60} \text{ h}$$

So, required distance travelled by car

$$= \text{speed} \times \text{time}$$

$$= 45 \times \frac{140}{60} = 105 \text{ km}$$

35. (b) Given, $x = 3$ km/h, $y = 4$ km/h,

$$t_1 = 15 \text{ min} = \frac{15}{60} \text{ h}$$

$$\text{and } t_2 = 15 \text{ min} = \frac{15}{60} \text{ h}$$

$$\therefore \text{Required distance} = \frac{xy(t_1 + t_2)}{y - x}$$

[Rule 4]

$$= \frac{3 \times 4 \left(\frac{15}{60} + \frac{15}{60} \right)}{4 - 3}$$

$$= 12 \times \frac{30}{60} = 12 \times \frac{1}{2} = 6 \text{ km}$$

36. (c) If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then = $\frac{2xy}{x + y}$

[Rule 2]

37. (b) Let speed of the flow of water be v km/h and rate of sailing of sailor be u km/h.

Speed of sailor downstream

$$= (u + v) \text{ km/h}$$

[Rule 9]

Speed of sailor upstream = $(u - v)$ km/h

$$\text{Condition I } u + v = \frac{48}{8}$$

$$\Rightarrow u + v = 6 \quad \dots(i)$$

$$\text{Condition II } u - v = \frac{48}{12}$$

$$\Rightarrow u - v = 4 \quad \dots(ii)$$

On subtracting Eqs. (ii) from (i), we get

$$v = 1 \text{ km/h}$$

38. (c) Speed of train = 72 km/h

$$= 72 \times \frac{5}{18}$$

$$= 20 \text{ m/s}$$

Length of train = distance covered

$$x = 20 \times 15$$

$$x = 300 \text{ m}$$

Length of the train is 300 m.

39. (b) Let the distance between two points be x km

Time taken by A - Time taken by B = 2

$$\Rightarrow \frac{x}{30} - \frac{x}{45} = 2$$

$$\Rightarrow x \left[\frac{3 - 2}{90} \right] = 2$$

$$\Rightarrow x = 90 \times 2 = 180 \text{ km}$$

40. (b) Let t h be the time taken by the man to reach his office at speed of 15 km/h.

Then, time taken to reach office at the speed of 10 km/h = $(t + 2)$ h

$$\text{Now, } s_1 \times t_1 = s_2 \times t_2$$

$$\Rightarrow 10 \times (t + 2) = 15 \times t$$

$$\Rightarrow 10t + 20 = 15t \Rightarrow 5t = 20$$

$$\therefore t = 4 \text{ h}$$

Now, distance covered to reach office

$$= s_1 \times t_1$$

$$= 10 \times (4 + 2) = 10 \times 6 = 60 \text{ km}$$

Now, speed required to reach office at 12 noon in 5 h.

$$\therefore \text{Speed} = \frac{60}{5} = 12 \text{ km/h}$$

41. (b) Speed of train = 48 km/h

$$= \left(48 \times \frac{5}{18} \right) \text{ m/s}$$

Let the length of train be x m

$$x = 48 \times \frac{5}{18} \times 9$$

$$x = 120 \text{ m}$$

Length of the train is 120 m.

42. (d) Let the speed of a train be x m/s and length be y m.

Condition I When $t = 10$ s

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$\Rightarrow 10 = \frac{y}{x} \therefore y = 10x \quad \dots(i)$$

Condition II When $t = 20$ s

and total distance = $y + 200$ m

$$\therefore \text{Time} = \frac{\text{Total distance}}{\text{Speed}}$$

$$20 = \frac{y + 200}{x} \Rightarrow 20x = y + 200$$

$$20 \times \frac{y}{10} = y + 200$$

$$\Rightarrow 2y - y = 200 \Rightarrow y = 200$$

Hence, the length of train is 200 m.

43. (b) Given, speed of a train = 40 km/h

$$= 40 \times \frac{5}{18} \text{ m/s}$$

Speed of another train = 20 m/s

∴ Required ratio

$$= \frac{\text{Speed of first train}}{\text{Speed of second train}}$$

$$= \frac{40 \times \frac{5}{18}}{20} = \frac{40 \times 5}{20 \times 18} = \frac{2 \times 5}{18}$$

$$= \frac{10}{18} = \frac{5}{9} \text{ or } 5:9$$

44. (a) Given, distance between two points (A and B) = 110 km

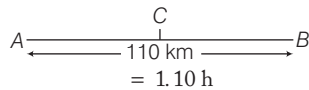
Their relative speed

$$= 60 + 40 = 100 \text{ km/h}$$

Time after which they meet

$$= \frac{\text{Total distance}}{\text{Relative speed}}$$

$$= \frac{110}{100}$$



Distance covered by A in 1.1 h

$$= AC = 60 \times 1.1 = 66 \text{ km}$$

Remaining distance

$$= BC = 110 - 66 = 44 \text{ km}$$

∴ Required ratio = AC : BC

$$= 66 : 44 = 3 : 2$$

45. (c) Let the total distance be d .

Time taken to cover $\frac{1}{3}$ rd distance,

$$t_1 = \frac{\frac{1}{3}d}{x} = \frac{d}{3x} \left[\because \text{time} = \frac{\text{distance}}{\text{speed}} \right]$$

Remaining distance = $d - \frac{1}{3}d = \frac{2}{3}d$

Time taken to cover $\frac{2}{3}$ rd distance,

$$t_2 = \frac{\frac{2}{3}d}{2y} = \frac{2d}{6y} = \frac{d}{3y}$$

Time taken to cover distance from A to B and B to A,

$$\therefore t = \frac{2d}{6z} = \frac{d}{3z}$$

According to the question,

$$t_1 + t_2 = t \Rightarrow \frac{d}{3x} + \frac{d}{3y} = \frac{d}{3z}$$

$$\Rightarrow \frac{1}{3x} + \frac{1}{3y} = \frac{1}{3z} \therefore \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

46. (a) Let the original speed of an aircraft be x km/h and its reduced speed = $(x - 200)$ km/h.

Condition I

Time taken by aircraft to cover 600 km = $\frac{600}{x}$ h.

Condition II Time taken by aircraft to cover

$$600 \text{ km} = \frac{600}{(x - 200)} \text{ h}$$

According to the question,

$$\frac{600}{x - 200} - \frac{600}{x} = \frac{1}{2} \left[\because 30 \text{ min} = \frac{1}{2} \text{ h} \right]$$

On dividing both sides by 600, we get

$$\frac{1}{x - 200} - \frac{1}{x} = \frac{1}{600 \times 2}$$

$$\Rightarrow \frac{x - (x - 200)}{x(x - 200)} = \frac{1}{1200}$$

$$\Rightarrow x^2 - 200x - 240000 = 0$$

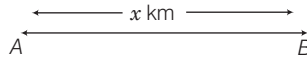
$$\Rightarrow (x - 600)(x + 400) = 0$$

$$\Rightarrow x = 600 \text{ km/h}$$

$$\text{or } x \neq -400 \text{ km/h}$$

$$\therefore \text{Required time} = \frac{600}{600} = 1 \text{ h.}$$

47. (b) Let the distance between A and B be x km and speed be V km/h.



Case I Given, distance = x km, speed = V km/h and time = 8 h.

$$\therefore \text{Speed} = \frac{\text{Distance}}{\text{Time}} \Rightarrow V = \frac{x}{8} \dots (i)$$

Case II If speed = $(V + 4)$ km/h

$$\text{and time} = 7\frac{1}{2} \text{ h} = \frac{15}{2} \text{ h}$$

$$\text{then, } V + 4 = \frac{x}{15/2} \Rightarrow V + 4 = \frac{2x}{15}$$

$$\Rightarrow \frac{x}{8} + 4 = \frac{2x}{15} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \frac{2x}{15} - \frac{x}{8} = 4 \Rightarrow \frac{x}{120} = 4$$

$$\Rightarrow x = 480 \text{ km}$$

48. (a)

Let total distance of AB be x km.

$$\text{For distance AC, } t_1 = \frac{x/3}{10} = \frac{x}{30} \text{ h}$$

$$\text{For distance CD, } t_2 = \frac{x/3}{20} = \frac{x}{60} \text{ h.}$$

$$\text{For distance BD, } t_3 = \frac{x/3}{60} = \frac{x}{180} \text{ h.}$$

∴ Total time taken

$$= t_1 + t_2 + t_3 = \frac{x}{30} + \frac{x}{60} + \frac{x}{180}$$

$$= \frac{10x}{180} = \frac{x}{18}$$

$$\therefore \text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}} = \frac{x}{\frac{x}{18}} = 18 \text{ km/h}$$

49. (c) Let speed of a boat in still water be x km/h and speed of the stream be y km/h.

Then, speed of the boat downstream

$$= (x + y) \text{ km/h} \quad [\text{Rule 9}]$$

and speed of the boat upstream

$$= (x - y) \text{ km/h}$$

According to the question,

$$\frac{32}{x + y} = 6 \quad \left[\because \text{speed} = \frac{\text{distance}}{\text{time}} \right]$$

$$\text{and } \frac{14}{x - y} = 6$$

$$\Rightarrow 6x + 6y = 32 \quad \dots (i)$$

$$\text{and } 6x - 6y = 14 \quad \dots (ii)$$

On subtracting Eq. (ii) from Eq. (i), we get

$$12y = 18$$

$$y = \frac{18}{12} = 1.5 \text{ km/h}$$

50. (a) Let the speed of B be x m/s.

$$\therefore \text{Speed of A} = 1\frac{2}{3}x = \frac{5x}{3} \text{ m/s}$$

Ratio of speed of rates of A and B

$$= \frac{5x}{3} : x = 5 : 3$$

∴ 2 m are gained in a race of 5 m.

∴ 1 m are gained in a race of $\frac{5}{2}$ m.

So, 80 m are gained in a race of

$$\frac{5}{2} \times 80 \text{ m} = 200 \text{ m}$$

51. (a) Given, speed of thief = 10 km/h

$$= \frac{10 \times 1000}{60} \text{ m/min} = \frac{500}{3} \text{ m/min}$$

and speed of policeman = 11 km/h

$$= \frac{11 \times 1000}{60} \text{ m/min} = \frac{550}{3} \text{ m/min}$$

Now, distance travelled by thief in 6 min

$$= \frac{500}{3} \times 6 = 1000 \text{ m}$$

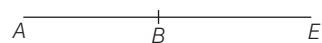
and distance travelled by policeman in

$$6 \text{ min} = \frac{550}{3} \times 6 = 1100 \text{ m}$$

∴ Difference = $(1100 - 1000) = 100 \text{ m}$

Hence, the distance between them after 6 min is 100 m.

52. (b) Let A and B will meet after t h at point E.



Distance travelled by A = pt h

and distance travelled by B = qt h

According to the question,

$$pt = qt + c \Rightarrow pt - qt = c$$

$$\Rightarrow t(p - q) = c \Rightarrow t = \frac{c}{p - q} \quad \dots (i)$$

∴ Distance travelled by A = pt

$$= \frac{pc}{p - q} \quad [\text{from Eq. (i)}]$$

53. (c) Let the original speed of car be x km/h.

$$\text{Time taken to cover 300 km} = \frac{300}{x} \text{ h}$$

If the speed of car is increased by 15 km/h.

$$\text{Then, } \frac{300}{x+15} = \frac{300}{x} - 1$$

$$\Rightarrow 1 = \frac{300}{x} - \frac{300}{x+15}$$

$$= 300 \left[\frac{x+15-x}{x(x+15)} \right]$$

$$\Rightarrow 1 = \frac{4500}{x^2 + 15x}$$

$$\Rightarrow x^2 + 15x - 4500 = 0$$

$$\Rightarrow (x+75)(x-60) = 0$$

$$\therefore x = 60 \text{ km/h}$$

54. (b) Total length of trains

$$= 121 + 99 = 220 \text{ m}$$

Relative speed of trains $= (40 + 32)$ km/h

$$= 72 \text{ km/h}$$

$$= 72 \times \frac{5}{18} \text{ m/s} = 20 \text{ m/s}$$

$$\therefore \text{Time} = \frac{\text{distance}}{\text{speed}} = \frac{220}{20} = 11 \text{ s}$$

55. (d) Let the distance be x .

Now, ratio of time taken to travel the distance by each bus is

$$\frac{x}{2} : \frac{x}{3} : \frac{x}{4} \Rightarrow \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$$

$$= 6 : 4 : 3$$

56. (a) Speed of faster train is 1.5 km/min.

i.e. 90 km/h or 25 m/s

and speed of slower train $= 60$ km/h

$$= \frac{50}{3} \text{ m/s}$$

Since, these trains are moving in same direction.

So, relative speed of train

$$= \left(25 - \frac{50}{3} \right) \text{ m/s} = \frac{25}{3} \text{ m/s}$$

Time taken by crossing slower from

$$= 27 \text{ s}$$

$\therefore$ Distance $= \text{speed} \times \text{time}$

$$= \frac{25}{3} \times 27 = 225 \text{ m}$$

57. (b) When A covers 100 m, then B covers 96 m and C covers 98 m.

i.e. when C covers 98 m, then B covers 96 m.

$\therefore$ when C covers 100 m, then B covers 100×96

$$\frac{98}{98} \text{ m, i.e. } 97.96$$

So, C beats B by approximately 2.04 m.

58. (a) Ratio of speeds are 16 : 15 : 11.

Let speed of winner athletic be $16x$ km/h.

Similarly, speed of athletic in second and third position be $15x$ and $11x$, respectively.

Total distance travelled by winner is 4 km.

$$\therefore \text{Time} = \frac{4}{16x} \text{ h}$$

Distance travelled by second athletic in

$$\frac{4}{16x} \text{ h} = \frac{4}{16x} \times 15x = \frac{15}{4} \text{ km}$$

Similarly, distance travelled by third athletic in

$$\frac{4}{16x} \text{ h} = \frac{4}{16x} \times 11x = \frac{11}{4} \text{ km}$$

$\therefore$ Difference between their distances

$$= \left(\frac{15}{4} - \frac{11}{4} \right) \text{ km}$$

$$= 1 \text{ km} = 1000 \text{ m}$$

59. (b) Refer to example 12.

60. (d) Petrol consumed by the bike

$$= (10 \times 20) + (30 \times 40)$$

$$+ (10 \times 20) \text{ mL}$$

$$= (200 + 1200 + 200) \text{ mL}$$

$$= 1600 \text{ mL} = 1.6 \text{ L}$$

61. (d) Refer to question 53.

62. (c) Let the speed of the train $= x$ km/h

Then, relative speed of train

$$= (x + 5) \text{ km/h}$$

and length of the train

$$= 225 \text{ m} = 0.225 \text{ km} \quad [\text{given}]$$

Time taken by train to cross the man

$$= \frac{0.225}{x+5} \text{ h}$$

According to the question,

$$\frac{0.225}{x+5} = \frac{9}{3600}$$

$$\Rightarrow x + 5 = 90$$

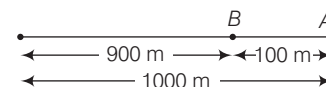
$$\Rightarrow x = 85 \text{ km/h}$$

Hence, the speed of the train is 85 km/h.

63. (c) Refer to question 46.

64. (a) Since, either A beats B by 100 m or 10 s. It means that B runs 100 m in 10 s.

$$\therefore \text{Speed of } B = \frac{100}{10} = 10 \text{ m/s}$$



$\therefore B$ gets injured at a distance of 450 m and his speed gets halved.

So, time taken by B to cover 1000 m

$$= \frac{450}{10} + \frac{550}{5} = 155$$

$\therefore$ Ratio of speed of A and B is equal to ratio of distance covered by A and B .

$$A : B = 1000 : 900 = 10 : 9$$

$$\text{Now, speed of } A = \frac{10}{9} \times 10 = \frac{100}{9} \text{ m/s}$$

Time taken by A to cover

$$1000 \text{ m} = \frac{1000}{\frac{100}{9}} \times 9 = 90 \text{ s}$$

Hence, A beat B by length of time

$$= (155 - 90) = 65 \text{ s}$$

65. (c) I. Let the length of race be x m.

Then, distance covered by $A = x$ m.

Distance covered by B when A reaches the destination $= x - 30$

Distance covered by C when A reaches the destination $= x - 48$

$$\text{Ratio of speed of } B \text{ to } A = \frac{x-30}{x}$$

$$\text{Ratio of speed of } C \text{ to } A = \frac{x-48}{x}$$

and distance covered by C when B reaches the destination $= x - 20$

$$\text{Ratio of speed } B \text{ to } C = \frac{x}{x-20}$$

$$\frac{x-30}{x} = \frac{x-48}{x-20} \left[\frac{B}{A} = \frac{C}{A} \times \frac{B}{C} \right]$$

$$\Rightarrow x^2 - 50x + 600 = x^2 - 48x$$

$$\therefore x = 300 \text{ m}$$

II. The speeds of A, B and C are in the ratio

$$300 : 270 : 252 = 50 : 45 : 42$$

07

TIME AND WORK

Usually (2-4) questions have been asked from this chapter. Generally questions are asked on Man-hours formulae, total wages paid for a work, number of days to complete a certain work and pipes and cisterns.



In this chapter, we will study the relationship among the quantity of work given, wages given, stipulated time, number of persons etc., and after it, we will be able to calculate the work in stipulated time by arranging some persons according to the work but before that let's discuss some basic rules.

IMPORTANT RULES AND FORMULAE

Rule 1 If a person can do a piece of work in ' n ' days, then he will do $\frac{1}{n}$ of the work in one day and if a person can do $\frac{1}{n}$ th of work in one day, then he will complete the work in n days.

e.g. If Raj can do a piece of work in 20 days, then he will do 1/20th of the work in one day.

Note In the problems related to time and work it is always considered that a man/woman works at uniform rate.

Rule 2 A and B can do a piece of work in x days and y days, respectively. Then, time taken by $(A + B)$ to complete the work is equal to reciprocal of $(A + B)$'s one day's work, i.e. $\frac{xy}{x + y}$.

Rule 3 If A and B can complete a work in x days and A alone can finish that work in y days, then number of days required to complete the work by $B = \frac{xy}{y - x}$ days

Rule 4 If A and B can do a piece of work in x days, B and C can do same work in y days, C and A can do same work in z days. Then, they will complete the same work in $\frac{2xyz}{xy + yz + zx}$ days by working together.